

Dynamic-Payload Pick-and-Pour Control for a UR5 Barista Robot: Progress Report

Soumaya El Mansouri, Jingxuan Yang, Tracy Yang
RBE 501, Worcester Polytechnic Institute
Worcester, MA, USA

I. INTRODUCTION

A. Problem Description

B. Related Work

Our task combines three problems that the literature has mostly treated separately: modelling the arm itself, perceiving how much liquid has been poured, and deciding how to pour. We organize prior work around these problems and, for each paper, state what we take from it and what we change.

UR5 modelling. Kebria et al. [1] derive the DH parameters, closed-form forward and inverse kinematics, and the Lagrangian dynamics (inertia, Coriolis, and gravity terms) of the UR5, and they verify the model in SimMechanics with a PD controller using feedback linearization. This is the most direct foundation for our project. Using these equations, we compute the Jacobian, manipulability, and joint torques along the pick-and-pour trajectory, and we use their DH table and inverse-kinematics as our starting point. The limitation is that their dynamics assume constant link and payload parameters. In our task, the arm holds a container that loses mass throughout the pour, so a controller built on their model with a fixed payload would see a changing gravity and inertia load as an unmodelled disturbance.

Vision-based liquid volume estimation. Schober et al. [2] estimate the liquid volume in transparent containers from a single RGB image and then execute a pouring motion selected from a simulation on a UR5e. Their pouring strategy of rotating the container about a fixed liquid exit point is useful for the small opening of a coffee container. Their results, however, show why vision alone is not enough for our task: the complete pipeline had RMSE of 21–26 mL for 30–50 mL targets, and the authors note that volume-estimation and simulation errors add up so the setup is not precise enough for laboratory use. Because a pour is selected before it starts and then executed without feedback, an error in the volume estimate cannot be corrected during the pour. We therefore use the camera only to localize the mug and container for grasping and placing, and we take the poured amount from a scale, which gives a direct measurement that can be used to stop the pour.

Force/torque perception of liquids. Luo et al. [3] use a wrist-mounted force/torque sensor on a UR5e to push a whisk through pancake batter and infer its uniformity, liquid level,

and water-flour ratio from the resistance torque. They then pour by rotating the wrist at a constant angular speed and use a learned model to set the arm speed from the batter properties. This matters for us for two reasons. First, it shows that a wrist force/torque sensor on this class of arm gives measurements precise enough to be useful (liquid level within about 3–4% error). Second, it uses the sensor at the same location we plan to, which supports using it to track the weight of the container in the gripper as it empties. Their pour, however, is open loop with respect to the amount poured: the properties are measured before pouring, and nothing is measured during the pour to decide when to stop. We close this loop by feeding the sensor and scale measurements back into both the stopping decision and the controller.

Learned pouring policies. Babaian et al. [4] train PourNet, a reinforcement-learning policy with curiosity-based exploration and curriculum learning in simulation, and execute its motion on a Kinova Movo with a nonlinear model predictive controller. It is the work closest to ours in goal. Its second arm holds the target container and measures its weight with a force/torque sensor, and the pour is based on this feedback, with deviations of a few grams across liquids. It also gives a practical recipe for the measurement itself using an IMU for gravity compensation and a low-pass filter on the force/torque signal (1 Hz cut-off), which we can adapt for our scale and wrist sensor. The cost is that the policy needs millions of simulated training steps and treats the pour as end-effector planning, so it does not model how the arm’s own dynamics change as the held container loses mass. For a course project with limited simulation time, a model-based controller whose structure is known and which can be analyzed with the Jacobian and manipulability tools from class is a better fit than a trained policy.

Comparison and our approach. Table I compares these works with ours. Taken together, the prior work establishes (i) a complete UR5 model [1], (ii) that vision alone is too imprecise for the poured amount [2], (iii) that wrist force/torque sensing carries useful information about the liquid [3], and (iv) that weight feedback from the target container gives accurate pours [4]. What none of them does is model the arm carrying a payload whose mass changes during the pour and design the controller around it. Our work fills this gap: we extend the UR5 model of [1] with a time-varying payload, combine

TABLE I
COMPARISON OF RELATED WORK AND THIS PROJECT

Work	Sensing	Pour control	Time-varying load modelled
[1]	–	PD tracking	No
[2]	RGB camera	Simulation-selected, no feedback	No
[3]	Wrist F/T	Open-loop	No
[4]	F/T (weight of target)	RL + NMPC	Learned implicitly
Ours	Camera, scale, wrist F/T	Model-based, dynamic load	Yes

a camera (object localization), a scale (poured mass), and a wrist force/torque sensor (payload mass) for feedback, and use a model-based controller that compensates for the changing load. We evaluate this on the complete pick-up-and-pour sequence in ROS2 and Gazebo and examine manipulability and singularities along the trajectory.

II. METHODOLOGY

A. Robot Sequence

B. Kinematic Model

C. Dynamic Model with Time-Varying Payload

D. Controller

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